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Establishing a Biostatistical Core Unit in a Clinical Research Center

STEPHAN ARNDT and ROBERT F. WOOLSON*

We have recently had the rare opportunity to open the door on a Biostatistical Core Unit within a Mental Health Clinical Research Center specializing in the multidisciplinary inquiry of schizophrenia. Based on the literature, our previous consulting experience, and the experience of colleagues, we implemented policies and procedures that avoided common problems facing consulting organizations and maximized effective collaboration between researchers and statisticians. This is an example of the application of policies, which involved assigning high priority to education, making ourselves available for consultation and collaboration, and developing an environment that fostered concordant research.

KEY WORDS: Biostatistical consulting unit; Organizational structure; Statistical consulting.

The body of literature on statistical consulting has grown markedly within the last several years, reflecting an increased interest and mounting recognition that shared information on aspects of consulting is important to a science such as ours. There are numerous references available on statistical issues in consulting, including books by Box, Hunter, and Hunter (1978), Cochran (1983), Ehrenberg (1978), Hoaglin, Mosteller, and Tukey (1983), Milliken and Johnson (1984), as well as other sources such as entire issues of *Biometrics*. Nonstatistical aspects of consulting have also been discussed in the statistical literature (e.g., Boen and Zahn 1982; Rustagi and Wolfe 1982; Zahn and Isenberg 1983) and in other fields (Gallessich 1983; O'Neill and Trickett 1982; Parsons and Meyers 1984). Despite this growing appreciation there seems to be few reported examples and little documentation on the problems and solutions in initiating a statistical consulting unit within a research organization such as a Clinical Research Center.

Recently, we have had the rare opportunity to open the door on a Biostatistical Core Unit in a Mental Health Clinical Research Center (MH-CRC) specializing in the study of schizophrenia. The literature, our previous consulting experience, and the experience of colleagues sensitized us to important considerations in developing the Biostatistical Core Unit. We have tried to implement policies and procedures that would tend to avoid common problems reported in consulting organizations and to maximize our effectiveness by providing an open and accessible environment for re-

searchers to collaborate among themselves and with statisticians.

ORGANIZATION OF THE BIOSTATISTICAL CORE UNIT

The Biostatistical Core Unit includes one full-time Ph.D. staff statistician, two other Ph.D.-level statisticians at 20% time, a full-time data manager, and two half-time people—a systems analyst and a secretary. This Core Unit's primary purpose is to provide biostatistical and data management consulting services to all MH-CRC investigators. The unit directly supports programming the core data base, which includes more than 2,000 variables and multiple data matrices taken from a variety of brain-scanning techniques. The Core Unit also maintains consulting support for the MH-CRC's Micro VAX II, which is a satellite node on a local area VAX cluster located in the campus computing center. We chose the location in a VAX cluster, in part, so that we could purchase operations, system, and cluster support from the university's computing center. This arrangement allows us to focus our services on providing biostatistical consulting.

To further allow us to concentrate on statistical collaboration, we have also spent a good deal of time developing documentation and training services on the use of the Micro VAX, interactive Statistical Analysis System (SAS), and the core data base. The time spent on this development and the maintenance of this philosophy throughout our functioning was an investment with potential for several benefits. Training coupled with pertinent readable documentation obviates much of the need for some kinds of consulting activity, especially questions dealing with simple computer-related problems. As a result, individuals can be made to feel more confident and may proceed with data-analytic activities with fewer technical irritations. Another consequence is the return on the initial time investment. The return is a reduction in the consultants' efforts spent on repeatedly answering questions about minor problems or matters and frees more time for interesting problems and independent investigation.

ORGANIZATION OF THE MH-CRC

The Biostatistical Core Unit operates within a larger research structure, and the nature, aims, and purposes of the larger center have an impact on the Core Unit's operations. The structure of the MH-CRC provided a favorable environment for development by establishing an independent core unit for biostatistical consulting and a unifying goal. The MH-CRC consists of seven research units (Diagnosis and Phenomenology, Brain Imaging, Genetics, Neuroendocrinology, Epidemiology, Basic Neurobiology, and Neu-

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ropharmacology) and three core units (Administrative, Assessment and Training, and Biostatistics). Because there is a unifying purpose to all of the units, that is, the study of the specific disease of schizophrenia, units typically operate in symbiotic relationships with one another. For instance, a study on response prediction and Haloperidol plasma levels in schizophrenia involves the Neuroendocrinology and Neuropharmacology Units with input from the Biostatistical Core Unit on design issues.

A characteristic that may distinguish our situation from other research organizations stems from the novelty of the research. Because formal hypotheses are premature or rudimentary, many investigators prefer simple descriptive analyses using visual inspection of data displays, frequency tables, graphics, etcetera. Investigators can perform many of these analyses by themselves with minimal reliance on the Biostatistical Core Unit. We encourage investigators to explore their own data by assigning high priorities to services that will support that behavior (e.g., education) while assigning lower priorities to doing analyses for them.

COMMON GOALS AND BROAD EXPOSURE

We found that the commonness of purpose, the study of a particular disease, greatly simplifies matters from the view of a statistical consultant, since there is a high degree of overlap in the variables used in these studies (e.g., indices of symptom severity). This allows a consultant to become well acquainted with the subject matter, many measurement nuances, and other properties, such as what to expect in terms of correlations and observed distributions. The consultant does not need to spend as much time learning about what to expect and can quickly spot abnormalities. This concentration of focus is quite different from a situation where there is a diffuse range of subject matter for the consultant to become acquainted with, each with its own independent set of methods, measures, goals, problems, and concerns. The development involved in establishing our biostatistical consulting unit was thus greatly simplified and would have required a very different approach if our intent had been to service multiple departments and a potpourri of goals.

The overall structure of the MH-CRC, which includes a central Biostatistical Core Unit and a variety of research units, has another benefit for the consultant. Our involvement in numerous projects and protocols allows exposure to problems that are pervasive throughout numerous studies yet may appear relatively minor within a narrower context. Thus problems with a widespread impact are especially visible, facilitating the possibility for what Marquardt (1981) called "total involvement" and suggesting solutions at various organizational levels, from computer program changes to study design or data management policy. For instance, repeatedly seeing symptom measures that look vaguely bimodal might suggest that what may appear to be a singular diagnostic group is in fact a mixture, or that there are stages of symptomatology, or perhaps that the diagnostic criteria are not being reliably followed, suggesting that additional training for raters may be required. Hands-on experience in the analysis of an instrument for one study may lead to suggested changes or cautions for a proposed study.

An example, which came up within the last year, indicates the possible seriousness of an overall problem that might easily be overlooked when seen within a particular study. Many of our data files contain very sensitive information. For any particular data set or data base, care has been taken so that there is no uniquely identifying information for an individual. However, the information from one protocol, coupled with the MH-CRC core data base and the magnetic resonance imaging data, would have readily yielded enough detail to pose a potential threat to confidentiality. Fortunately, the situation was easily and quickly averted by omitting unnecessary variables. At this time we also tightened the security of our data files. Although similar situations may present difficulties in all such large clinical research centers, they highlight the necessity for total involvement in the spectrum of studies by the consulting statistician.

Fostering close involvement and accessibility for collaboration produces a particular kind of client-consultant relationship within the organization. Functionally, the individuals in the Biostatistical Core Unit have the advantage of being internal consultants, tending to be seen by investigators as convenient and easily approachable collaborators in research. Decisions about design or analysis are shared rather than dictated, although care is taken to explain the pros and cons of various options. Every attempt is made to include all of the parameters of the design and analysis problem, including the audience and investigator, in finding at least a mutually acceptable protocol. We emphasize and prefer the sharing of educated decisions about the methodological alternatives and their limits of meaningful inference. We feel that this attitude is congruent with our overall goal of providing education and collaboration. If researchers choose to pick what we believe to be a less than optimal solution, then we hope that they do so fully cognizant of the problems and the alternate available paths. Whenever a formal review of a protocol is made, alternative methods are described in writing and include the limitations, benefits, and our suggested strategy. These written reviews are then sent to all involved parties.

Although in-house consultation does have a large number of benefits in terms of convenience and mutual substantive and personal familiarity, Gibbons and Freund (1980), in their review of organizational structures for consulting within universities, noted many unsatisfactory aspects. One problem that they felt might occur results directly from restricted communication between the in-house consultant and other biostatisticians, leading to a narrowness of techniques suggested and, possibly, less than optimal solutions to statistical problems. Gibbons and Freund (1980) also mentioned the problem of relatively minor (although perhaps frequent) recognition for the consultant (e.g., junior authorship or acknowledgments), which does little for professional prestige, tenure, etcetera. Most of these difficulties stem from the physical and structural isolation of the consultant from other statisticians who might provide collaboration, advice, and peer support and recognition.

We have avoided some of these problems through formal connection to an academic Division of Biostatistics within the university. As initially proposed, the Biostatistical Core Unit is a distinct entity within the MH-CRC, reporting to

the Director of the MH-CRC as well as to the Director of the Division of Biostatistics. Although funds for personnel in the unit come from the MH-CRC, formal appointments are made in the Division of Biostatistics, Department of Preventive Medicine and Environmental Health. Another consequence of this situation is that evaluation of the Biostatistical Core Unit is conducted by individuals in both the MH-CRC and Division of Biostatistics. The Division of Biostatistics is used to similar relationships and has been extremely supportive of the Core Unit's needs and difficulties. The association with the Division of Biostatistics gives the Core Unit an initial credibility, as several MH-CRC investigators had previously worked with or received formal training in the division. Thus the Biostatistical Core Unit is afforded a certain amount of structural independence and integrity, not unlike that seen in an independent statistical consulting center, plus a close tie to other biostatisticians. Two issues do, however, arise in this situation—the degree of academic involvement and the lack of financial independence.

THE RELATIONSHIP OF THE BIOSTATISTICAL CORE UNIT TO OTHERS

To the extent that we are in-house, we are within the MH-CRC in one sense, and yet we are attached to the academic Biostatistics Division. As such, the activities of our Biostatistical Core Unit have little or no direct bearing on students in biostatistics, our direct educational influence being directed to researchers (faculty and students) involved with the MH-CRC. We also do not, as yet, offer biostatistics students a platform for consulting experience in the MH-CRC. Students in biostatistics do receive consulting experience in the Biostatistics Division General Consulting Center. However, those individuals who are connected with programs in both Biostatistics and Psychiatry receive the educational consulting, assistance, and collaboration furnished by the Biostatistical Core Unit. Some of these people are in a work relationship with projects in the Department of Psychiatry and do, from time to time, receive informal consulting on projects. Other individuals are involved in postdoctoral programs combining psychiatry and biostatistics or epidemiology.

The Biostatistical Core Unit is also a research unit. Establishing an initial credibility through our association with the Division of Biostatistics, as well as demonstrating quality and efficient support for biostatistics in the MH-CRC, set the groundwork for our being perceived as an active research core that, like the other units, is in a collaborative situation. A major goal of a CRC is to promote research and independent research of an applied statistical nature resulting in published reports in respected journals in accordance with goals of the MH-CRC. Under these conditions we have not found it difficult to promote the idea that methodological advances can also be of use to MH-CRC researchers in their own endeavors or to ask them to collaborate with us.

As noted, monies for the Biostatistical Core Unit are derived from the MH-CRC grant. With the exception of the salaries for the 100%-time statistician and the 50%-time systems analyst, which are paid to the Department of Pre-

ventive Medicine's Biostatistical Division, all monies for the Core Unit come from an MH-CRC general fund controlled by the director of the CRC. Thus the Core Unit has no control over finances and must make personal requests to the director to receive funds for any expenditures. To date, this has not proved to be a problem, since most requests have been granted, with the exception of adding professional staff. In our case, we simply outlined what could and could not be accomplished without the additional help.

The fact that all monies are held in an MH-CRC controlled general fund could affect the priority attributed to the CRC project by the Department of Preventive Medicine Division of Biostatistics, since to some extent departmental priorities are related to economic issues. This would tend to force the MH-CRC and the Biostatistical Core Unit to become more self-reliant since any services they receive from the Department of Preventive Medicine are gratis and motivated by professional interest. The existence of an MH-CRC general fund also puts our unit into competition with other units for an already-tight budget. As a result we cannot freely provide some services, such as programming and data-analytic support. On the positive side, it compelled us to streamline, prioritize, and strive for efficiency.

SETTING EXPECTATIONS

We observed that many of the problems arising with consultation that have profound effects on outcome and morale stem from undefined expectations, and misunderstandings of responsibility. Therefore, we believed that it was crucial to define goals, purposes, and what researchers could and could not reasonably expect of the Core Unit early on and in as positive a way as possible. We also realize that the definition of expectations is an evolving process. Two aims were specified in the grant proposal—one, to provide statistical consultation to projects connected with the MH-CRC; and two, to provide computer support for MH-CRC core data. Other goals mentioned in the grant include quality control of the core data and feasibility evaluations of projects, as well as statistical research projects. More specific goals for the Biostatistical Core Unit were stated in a document titled "Services of the Biostatistics Core Unit of the MH-CRC." This statement was distributed within the first months of operation to all MH-CRC staff, investigators, and individuals who may potentially become involved with the research here. This was done to formally define our responsibilities and boundaries.

The Biostatistical Core Unit would contribute methodologic services to individual projects in the following areas:

1. Specifying research hypotheses in more specific and statistical terms.
2. Experimental design.
3. Reviewing the formalization of procedures; for example, definitions of the patient populations, randomization procedures, masking or blinding procedures, rule for inclusion and exclusion of patients.
4. Choice of statistical analysis strategies and alternatives.
5. Reliability assessment of measurements.
6. Determination of sample size.

7. Assistance with development of data forms.
8. Reviewing statistical portions of grants and/or papers.
9. Interpretation and presentation of statistical results.

Further to make efficient use of the biostatistical services, it recommended that investigators involve the core unit at all stages where statistical input might be required. In those situations where the unit has been involved in the preparation of a grant proposal or submission of a paper, the staff of the unit would be prepared to assist with presentations and defense of statistical matters at site visits or at professional meetings. A member of the Biostatistical Core Unit would also serve on the committee responsible for review of MH-CRC protocols.

Although the Biostatistical Core Unit is a free service to MH-CRC-connected investigators, motivation for documenting our list of activities somewhat parallels Boen's (1982) establishment of a rigid pricing structure for his Biostatistical Consulting Laboratory at the University of Minnesota, in that it attempts to predefine and solidify expectations. Once set, expectations are frequently difficult to modify. The explicit and formal documentation of our activities early in our development helped to define the expectations that might otherwise have developed haphazardly from others' preconceived notions about our responsibilities.

Even in the context of neatly defined expectations and priorities, emergency situations arise that demand our consideration and, at times, our flexibility. These might include analysis for a deadline forgotten by a researcher or a researcher's wish to change a form and consequent data entry programs and to begin collecting data using the new version immediately. One observation is that, although many emergencies are mainly only perceived crises, most are at worst counterproductive and at best disruptive to the efficient operation of the MH-CRC as a whole. Close involvement by the statistician throughout a project has often prevented emergencies that have a bearing on the Biostatistical Core Unit from occurring. In spite of every attempt at prevention, emergency situations do occur and challenge our operations. There is a crucial need for communication and negotiation at these junctures. The fact that these emergency events disrupt the entire MH-CRC must be made clear to all concerned parties. Furthermore, it is outlined what activities the Biostatistical Core Unit must desert to deal with the new request for immediate service. Frequently, in this context, it becomes obvious that the new request is far less central to the MH-CRC's goals than originally thought; however, whenever there is doubt, the decision of priorities is ultimately left to the MH-CRC Director. We also include at the beginning of the negotiations for immediate service a collaborative effort to go to any lengths in preventing a similar emergency from occurring again. Frequently, these preventive measures simply require additional lines of communication.

EVALUATIVE CRITERIA OF THE BIOSTATISTICAL CORE UNIT

Of course, goals of the Biostatistical Core Unit must be related to the evaluative criteria used for its performance.

Using Marquardt's criteria (1981) for evaluating statistical consulting in industry as an ideal, a major goal of the Biostatistical Core Unit is to increase the quality and productivity of research at the MH-CRC. More specific criteria are used to evaluate the performance of the staff statistician since, as the only full-time employee, his hours comprise the majority of hours spent on unit work. In some sense the criteria used to evaluate his performance can in part apply to the unit's performance. A synopsis of the performance standards on file for the staff statistician are given as follows:

1. Attendance and involvement in standing MH-CRC meetings where statistical consultation is of value to the MH-CRC as a whole or the Biostatistical Core Unit requires representation (weekly staff meetings, Investigators Meeting, Project Review Meetings, mock site visits).
2. Provide timely and up-to-date education, training sessions, documentation, and technical reports dealing with the MH-CRC VAX, SAS, and statistical procedures of general interest for MH-CRC personnel as needed.
3. Be available to investigators and MH-CRC personnel for informal in-house and formal biostatistical consultations on measurement issues, experimental design, development of instruments, analysis strategies and alternatives, interpretation, SAS-PROC's, and so forth. The consultation should involve close collaboration and thorough follow-up on the problem or project.
4. Complete constructive reviews and possible writing of statistical portions of grants and papers involving the MH-CRC. These reviews will be reported with recommendations promptly to all relevant individuals verbally. Once verbal feedback is received from the investigators, the reviews will be submitted in writing.
5. Provide general and specific collaborative contributions to research in biostatistics, psychiatric research, and methodology, which may be realized as a substantive contribution to individual projects or independent research of an original nature pertaining to MH-CRC research problems.
6. Involvement in, and contribution to, the development of protocols, data entry forms, research projects and hypotheses, grant proposals, etcetera. This involvement will be in the form of attendance at meetings and written and oral review.
7. Keeping up with developments in statistics, biostatistics, consulting, and the fields relevant to research at the MH-CRC.
8. Conducting independent biostatistical research of relevant data sets and developing innovative approaches (when appropriate) to MH-CRC data sets for which standard methods of analysis are not available (e.g., longitudinal data, neuroimaging data).
9. Writing up results of such independent research for publication or presentation at professional meeting.

These written statements (the foregoing list of evaluative objectives and the list of activities given in the previous section) serve to educate others as well as to advertise the Biostatistical Core Unit. Individuals may not make use of collaboration with the Biostatistical Core Unit or may have unrealistic expectations simply because of a lack of infor-

mation. Further, the documents serve as a resource to the staff to help them define their responsibilities.

Of the goals and activities mentioned so far, there is one area that is purposefully omitted from our lists of activities, that of performing routine data analysis and individual projects' data management. Clearly, the small staff of the Biostatistical Core Unit could not be responsible for the data-analysis needs of even a few productive researchers. As researchers sometimes expect such work, this can become a problem. On one hand, researchers need to have their data analyzed; on the other, the staff statistician cannot do it all. Additional problems arise when the core unit can do some but not all analytic tasks. Boen (1982) suggested that there should be absolutely no exceptions to his rigid pricing structure, citing a case where news of one exception was spread rapidly and suggesting that time was negotiable. This led to a situation allowing negotiation and a subsequent inability to decide which projects and clients were more worthy. Further, O'Neill and Trickett (1982) cited an example where an exception was seen as a challenge by the consultant but that her "temporary" role took on a rigidity that was impossible to change and ultimately led to her failure.

In an effort to provide services, yet not get bogged down with day-to-day analysis or simple programming, we took a three-pronged educational approach. The unit has invested a fair amount of time preparing, revising, and implementing a series of modular training sessions for investigators and staff on using our VAX computer, SAS, and, for those with previous SAS experience, a short course on SAS on the VAX. Throughout the training sessions we emphasize the added flexibility of interactive SAS. This provides an attractive alternative for investigators who want immediate results. In addition, we wrote a series of simple-to-follow "cookbook" manuals and made them available to allow researchers to get most simple analyses done quickly and without needing a programmer or a consultant. Finally, we have furnished many individuals with prototype programs with extensive documentation to get them started or supplied them with preprocessed data sets as SAS system files. Many of the queries investigators make of the data involve data displays, and the SAS procedures for these are straightforward. Researchers can be taught these techniques relatively quickly, becoming more self-sufficient and consequently reducing their demands on the Biostatistical Core Unit. During our first year, the vast majority of the systems analyst's and perhaps 20% of the staff statistician's time was devoted to these efforts.

Issues that involve maintaining and updating data sets have been addressed with a similar philosophy—that the Biostatistical Core Unit's primary concern is maintaining the quality of our core data base and supporting an environment for independent data bases. A primary aim was to provide a high quality set of data that would cut down on the necessity of ad hoc data cleaning and tracking down whether outliers represented entry errors or real measurements. As the MH-CRC core data base containing comprehensive broadly based information (e.g., demographics, diagnoses, patient histories) grows, and as progressively more research projects develop to the point of needing their own data bases, data management responsibilities will be-

come more complex. As researchers collect their own data to be coupled with selected variables in the core data base or generate subsamples, the number of specific data sets will rapidly become unmanageable. Since these individualized data sets can be modified at will by the researcher or staff, the Biostatistical Core Unit can in no way vouch for the data's structure or quality after such modifications. The small Biostatistical Core Unit cannot be responsible for individual data sets. Rather, the responsibility needs to be focused on qualities such as the integrity and accessibility of the core data and the ease of use, as well as consultation and help in designing the smaller data bases.

By establishing our limits of responsibility for data, we are setting up the expectations about the responsibilities of the individual research project. Since a member of the Biostatistical Core Unit reviews proposals that involve the MH-CRC, typically prior to submission, it is possible to make this expectation even more explicit in a written review. Usually all that is required is the suggested inclusion of a line item to cover some percentage of time for a data analyst, programmer, system analyst, coder, or other personnel. This inclusion accomplishes two goals. First, it makes it clear to the researcher that the Biostatistical Core Unit will not automatically furnish these services. Second, it assures that the researcher has the data-analytic capabilities to complete the project. The estimated amount of time involved is also based on the statistician's experience and judgment of the complexity of the full data-analytic task.

All in all we have had very good experience with the new unit. The environment within and surrounding the MH-CRC has provided the necessary ground for the development of our Biostatistical Core Unit. The location of the Unit as an in-house consulting unit, but which directly associates with an external Division of Biostatistics, gives us several major benefits and also helps us avoid some common problems (e.g., isolation from other statisticians or being seen as an outsider by investigators). Although we have seen specific activities of the Biostatistical Core Unit within the MH-CRC change quickly as different projects reach different stages of development, our attempts to explicitly formulate our role and expectations from the onset has also proved to be a successful strategy. Another tactic that has proved valuable is our emphasis on education. Education is frequently mentioned in many descriptions of what happens during a successful consulting session. Formalizing this in both our training sessions, literature and communication have proved to be highly effective means of fostering self-sufficiency and an environment supporting productive research for all parties. These strategies foster not only a successful consulting situation, but an actively collaborative arrangement, and promote the mutuality of the role a biostatistical core unit might play within a larger research organization.

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